

# softstart\_\_routed

## Design Report

kicad-tools 0.13.0

Rev 1 | 2026-06-12 | jlcpcb-tier1

### Board Summary

| Property   | Value                                 |
|------------|---------------------------------------|
| Layers     | 4 copper (F.Cu, In1.Cu, In2.Cu, B.Cu) |
| Footprints | 78 (56 SMD, 18 THT, 4 other)          |
| Nets       | 41                                    |
| Traces     | 4727 segments                         |
| Vias       | 158                                   |
| Board Size | 150.0 x 100.0 mm                      |

### Design Overview

#### Theory of Operation

Generator Soft-Start - Supercapacitor Power Assist

120VAC soft-start for 8000 BTU AC on Honda EU1000i

STM32G031K8T6 MCU, 2x30S supercap banks, back-to-back FETs + UCC27211 drivers

#### Power Architecture

**Power Rails:** +3V3, GND, PWR\_FLAG

| Regulator | Device      |
|-----------|-------------|
| U4        | XC6206-3.3V |
| U9        | LM7812      |

### Assembly Notes

1 fine-pitch component; 6 polarized components

- **Fine-pitch components:** 1 (U1)
- **Polarized components:** 6 – check orientation markings

## Off-board Assemblies

The following subsystems are part of the design but are **not placed on the PCB** – they connect through board connectors and are assembled by hand (DNP for fab assembly).

### SUPERCAP\_BANK\_POS

Positive half-cycle supercapacitor bank (off-board, 30S string)

| Property        | Value                              |
|-----------------|------------------------------------|
| Board connector | J3                                 |
| Part            | Tecate TPLH-2R7/12WR10X30          |
| Quantity        | 30                                 |
| Voltage         | 81V nominal                        |
| Capacitance     | 0.4F                               |
| Assembly        | hand solder (DNP for fab assembly) |

**Wiring:** J3 pin 1 -> SCAP\_POS+ (bank positive terminal); J3 pin 2 -> SCAP\_POS\_GND (bank return, star ground). 30 cells in series, 12F 2.7V each.

### SUPERCAP\_BANK\_NEG

Negative half-cycle supercapacitor bank (off-board, 30S string)

| Property        | Value                              |
|-----------------|------------------------------------|
| Board connector | J4                                 |
| Part            | Tecate TPLH-2R7/12WR10X30          |
| Quantity        | 30                                 |
| Voltage         | 81V nominal                        |
| Capacitance     | 0.4F                               |
| Assembly        | hand solder (DNP for fab assembly) |

**Wiring:** J4 pin 1 -> SCAP\_NEG+ (bank positive terminal); J4 pin 2 -> SCAP\_NEG\_GND (bank return, star ground). 30 cells in series, 12F 2.7V each.

## ERC Status

| Metric   | Count |
|----------|-------|
| Errors   | 0     |
| Warnings | 0     |

**Status:** SKIPPED – ERC skipped by user request

## Bill of Materials

| Value     | Package                                            | Qty | References                          | MPN |
|-----------|----------------------------------------------------|-----|-------------------------------------|-----|
| 100nF     |                                                    | 5   | C5, C20, C22, C23, C25              |     |
| 100nF     | C_0402_1005Metric                                  | 8   | C1, C2, C3, C8, C30, C31, C33, C34  |     |
| 10uF      |                                                    | 2   | C21, C24                            |     |
| 10uF      | C_0805_2012Metric                                  | 3   | C6, C7, C32                         |     |
| 4.7uF     | C_0805_2012Metric                                  | 1   | C4                                  |     |
| RB157     | Diode_Bridge_DIP-4_W7.62mm_P5.08mm                 | 1   | D1                                  |     |
| SMBJ18A   | D_SMB                                              | 4   | D_TVS1, D_TVS2, D_TVS3, D_TVS4      |     |
| STATUS    |                                                    | 1   | D2                                  |     |
| 15A       |                                                    | 1   | F1                                  |     |
| AC_INPUT  | TerminalBlock_bornier12_P5.08mm                    | 1   | J1                                  |     |
| AC_OUTPUT | TerminalBlock_bornier12_P5.08mm                    | 1   | J2                                  |     |
| SCAP_NEG  | TerminalBlock_bornier12_P5.08mm                    | 1   | J4                                  |     |
| SCAP_POS  | TerminalBlock_bornier12_P5.08mm                    | 1   | J3                                  |     |
| SWD-6     |                                                    | 1   | J5                                  |     |
| 2N7002    | SOT-23                                             | 2   | Q7, Q8                              |     |
| AO3400    | SOT-23                                             | 2   | Q5, Q6                              |     |
| IRFB4110  | TO-220-3_Vertical                                  | 4   | Q1A, Q1B, Q2A, Q2B                  |     |
| 100R      | R_Axial_DIN0617_L17.0mm_D6.0mm_P25.40mm_Horizontal | 12  | R20, R21                            |     |
| 10k       |                                                    | 7   | R2, R10, R11, R23, R24, R26, R28    |     |
| 10k       | R_0805_2012Metric                                  | 6   | R_GB1, R_GB2, R_GB3, R_GB4, R5, R29 |     |
| 150R 5W   | R_Axial_DIN0617_L17.0mm_D6.0mm_P25.40mm_Horizontal | 1   | R6                                  |     |
| 1k        |                                                    | 2   | R12, R22                            |     |
| 275VAC    | RV_Disc_D12mm_W4.2mm_P7.5mm                        | 1   | RV1                                 |     |
| 290.0k    |                                                    | 2   | R25, R27                            |     |
| 33k       | R_0805_2012Metric                                  | 2   | R3, R4                              |     |
| 5mR       | R_2512_6332Metric                                  | 1   | R9                                  |     |
| 990.0k    |                                                    | 1   | R1                                  |     |
| RESET     |                                                    | 1   | SW1                                 |     |
| H11AA1    | DIP-6_W7.62mm                                      | 1   | U2                                  |     |
| INA180A3  | SOT-23-5                                           | 1   | U3                                  |     |
| LM393     | SOIC-8_3.9x4.9mm_P1.27mm                           | 3   | U7, U7, U7                          |     |
| LM7812    | TO-220-3_Vertical                                  | 1   | U9                                  |     |
| MCP6001   | SOT-23-5                                           | 1   | U8                                  |     |

| Value                     | Package                                              | Qty | References        | MPN                       |
|---------------------------|------------------------------------------------------|-----|-------------------|---------------------------|
| STM32G031K8T6             | LQFP-32_7x7mm_P0.8mm                                 | 1   | U1                |                           |
| UCC27211                  | SOIC-8_3.9x4.9mm_P1.27mm                             | 2   | U5, U6            |                           |
| XC6206-3.3V               |                                                      | 1   | U4                |                           |
| Tecate TPLH-2R7/12WR10X30 | Off-board via J3 (hand solder, DNP for fab assembly) | 30  | SUPERCAP_BANK_POS | Tecate TPLH-2R7/12WR10X30 |
| Tecate TPLH-2R7/12WR10X30 | Off-board via J4 (hand solder, DNP for fab assembly) | 30  | SUPERCAP_BANK_NEG | Tecate TPLH-2R7/12WR10X30 |

## DRC Status

| Metric   | Count |
|----------|-------|
| Errors   | 0     |
| Warnings | 0     |
| Blocking | 0     |

**Status:** PASS

## Manufacturing Readiness

Verdict: READY

### Action Items

- [OPTIONAL] Verify zone fill in KiCad for 12 zone-connected nets
- [OPTIONAL] Analog-sensitive: U3 (INA180A3) — instrumentation amplifier (TI INA); manual layout review recommended
- [OPTIONAL] Analog net: AC\_LINE — audio signal; keep short, away from digital/switching nets
- [OPTIONAL] Analog net: BUS\_LINE — audio signal; keep short, away from digital/switching nets
- [OPTIONAL] Analog net: FUSED\_LINE — audio signal; keep short, away from digital/switching nets
- [OPTIONAL] Analog net: ISENSE\_NEG — analog signal; noise-sensitive, avoid crossing digital signals
- [OPTIONAL] Analog net: ISENSE\_POS — analog signal; noise-sensitive, avoid crossing digital signals
- [OPTIONAL] Analog net: I\_SENSE\_OUT — analog signal; noise-sensitive, avoid crossing digital signals
- [OPTIONAL] Analog net: V\_AC\_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- [OPTIONAL] Analog net: V\_AC\_SENSE\_RAW — analog signal; noise-sensitive, avoid crossing digital signals
- [OPTIONAL] Analog net: V\_BANK\_NEG\_SENSE — analog signal; noise-sensitive, avoid crossing digital signals
- [OPTIONAL] Analog net: V\_BANK\_POS\_SENSE — analog signal; noise-sensitive, avoid crossing digital signals

### Analog Components

1 analog-sensitive component detected – manual layout review recommended.

| Reference | Value    | Reason                             |
|-----------|----------|------------------------------------|
| U3        | INA180A3 | instrumentation amplifier (TI INA) |

## Routing Status

| Metric                | Value                 |
|-----------------------|-----------------------|
| Signal Net Completion | 100.0% (27/27)        |
| Overall Completion    | 100.0%                |
| Complete Nets         | 41 / 41               |
| Zone-Connected Nets   | 14                    |
| Single-Pad Nets       | 2 (no routing needed) |
| Incomplete Nets       | 0                     |
| Unconnected Pads      | 0                     |

### Zone-Connected Nets

- +3.3V
- AC\_LINE
- AC\_NEUTRAL
- BUS\_LINE
- FUSED\_LINE
- GND
- SCAP\_NEG+
- SCAP\_NEG\_GND
- SCAP\_POS+
- SCAP\_POS\_GND
- SRC\_NEG
- SRC\_POS
- VGATE
- VRECT

### Single-Pad Nets

2 single-pad nets (no routing needed) – not listed individually.

## Cost Estimate

| Metric                   | Per Board (estimated) |
|--------------------------|-----------------------|
| PCB Fabrication          | ~5.0 USD              |
| Components (estimated)   | ~5.72 USD             |
| Assembly (estimated)     | ~0.0 USD              |
| <b>Total (estimated)</b> | <b>~10.72 USD</b>     |
| Batch Quantity           | 5                     |
| Batch Total (estimated)  | ~53.6 USD             |